Analysis of Alkalinity of Seawater Using Titration and UV Spectroscopy

Introduction

The alkalinity of a solution is determined by the solution's ability to neutralize acids. Alkalinity also acts as a buffer, which resists pH changes when acids or bases are added to the water. In naturally occurring water, bicarbonate is the most common source of alkalinity due to the carbonate minerals that end up in the water from natural weathering of rocks. In seawater specifically, the alkalinity level is primarily dependent on the presence of carbonate, bicarbonate, and hydroxide ions, although carbonate and bicarbonate make up majority of basic species in seawater. Trace levels of alkalinity can also be a result of borates, phosphates, silicates, and other dissolved bases. Alkalinity in seawater is typically measured in parts per million calcium carbonate [3]. This number does not represent the total amount of calcium carbonate in the seawater, but the alkalinity of the seawater that corresponds to dissolving that amount of calcium carbonate in water. Since the alkalinity in seawater is made up of a combination of carbonate, bicarbonate, and some other bases, the total alkalinity of all of the bases in seawater corresponds to the hypothetical amount of calcium carbonate it would take to reach that same alkalinity in water. The ideal alkalinity level for seawater aquariums is between 143-214 ppm [5].

There are growing concerns over ocean acidification as a result of carbon dioxide dissolving into water and increasing acidity. In seawater, the typical pH level ranges from 7.5-8.5. Currently, seawater is at about 8.1 pH level [1]. When seawater is lower than 7.5, the acidity of the water can damage ecosystems. Many forms of marine life use carbonate to build shells and skeletons. In slightly acidic conditions, these animals cannot obtain the carbonate they need from the water to build their shells and skeletons [1]. As the water gets even more acidic, many of these carbonate structures will start to dissolve as the carbonate reacts with the hydronium ions in the water, increasing the risk of endangerment or extinction of many species.

In this experiment, I use a combination of pH titration with indicators and spectrophotometric determination of overall pH to analyze the concentration of carbonate and bicarbonate in a sample of seawater from a controlled fish habitat.

To measure the alkalinity of the solution, I decided to titrate the unknown alkalinity of seawater against a known concentration of hydrochloric acid solution. Since the ions that affect alkalinity are all basic, titrating with a known concentration of a strong acid will give me the amount of hydronium ions that are needed to neutralize the solution. Then, using the Henderson Hasselbach equation, I can determine the overall pH of the solution, given a measurement of concentration of either form of an indicator dye.

In my initial method brainstorming, I considered using phenolphthalein as an indicator. Phenolphthalein is often used as a pH indicator because in acidic solutions, it is colorless. At pH 8.3 and higher, it turns a vibrant pink. I planned on adding phenolphthalein to the sample of seawater, then titrating with hydrochloric solution until the pink color faded. However, at an equivalence point of 8.3, bicarbonate and hydroxide ions will be fully titrated, but carbonate ions will only be halfway neutralized into bicarbonate. This method will only be effective in telling me how much acid I need to titrate monoprotic bases but will not be effective in fully neutralizing the carbonate to give me an overall pH of the entire sample.

The next method I considered involves both a titration and a spectrophotometric determination of pH. First, I will use bromocresol green as my pH indicator for the titration. Bromocresol green changes from a bright yellow in acidic solutions to a dark blue in basic solutions with an equivalence point between 3.8 and 5.4 pH. Using a known concentration of hydrochloric acid, I will titrate until I reach an equivalence point. When titrating, I am looking for a green midpoint to signify that enough hydrochloric acid has been added to the sample of seawater to protonate the carbonate ions twice and fully neutralize the solution. The volume of hydrochloric acid used, multiplied by its known concentration, will be the number of moles of hydronium needed to fully titrate all basic ions, including carbonate.

There are 2 titration reactions happening. First, bicarbonate is being protonated one time to form carbonic acid. The other reaction is where carbonate needs to be protonated twice to form carbonic acid. At bromocresol green equivalence point at pH 3.8-5.4, both reactions are fully complete.

Titration from bicarbonate to carbonic acid: $HCO_3^- + H_3O^+ \rightarrow H_2CO_3$

Titration from carbonate to carbonic acid: $CO_3^{2-} + 2H_3O^+ \rightarrow H_2CO_3$

The number I obtain from this titration is the number of moles of hydrochloric acid it took to titrate both the bicarbonate and carbonate in the seawater. Recall that the alkalinity of seawater is typically reported as some hypothetical concentration of only carbonate it would take to reach the same level of alkalinity. Therefore, I would need to convert this number into an equivalent amount of just carbonate. If I assume that all the alkalinity in the seawater is a result of carbonate ions, I know that the molar ratio of carbonate to hydrochloric acid to neutralize the carbonate is 2:1. Therefore, I can divide the number of moles of hydrochloric acid by 2 to get the moles of hypothetical carbonate it would take to reach the same alkalinity. After converting from moles of carbonate to ppm of carbonate, this is my alkalinity of the seawater expressed in units of ppm of calcium carbonate in solution.

Then, I will use m-cresol purple dye to use spectrophotometric determination to analyze the overall pH of the seawater. M-cresol purple is a sulfonephthalein indicator that presents at different colors in different forms. The indicator has 3 main forms: H2I, HI-, and I2-, which appear pink, yellow, and purple respectively. After putting drops of the indicator in the I2- state into the seawater, the indicator will latch onto any remaining hydronium ions and transition to the color that corresponds to the newly protonated ion. Since each ion will absorb a different wavelength when hit with light, I can use a spectrophotometer to measure the absorbance of each sample at different wavelengths, which will then tell me the concentrations of each type of ion through Beer's Law. This will then tell me what the concentration of hydronium is in the seawater, allowing me to determine the overall pH of the seawater.

After obtaining both the overall pH of the seawater and the concentration of carbonate, I can determine the concentration of bicarbonate through the Henderson Hasselbach equation. Through these calculations, I plan on obtaining an overall pH of the seawater sample as well as the concentrations of carbonate and bicarbonate in the sample. These values will provide more insight into the seawater's buffering capacity based on its alkalinity measurements.

Results

The accepted level of alkalinity in seawater aquariums is between 143-214 ppm, and the recommended pH level is between 8.0-8.3. Day 0 is marked as the first day the fish arrived in the tank.

Day	pH	95% CI	Alkalinity (ppm)	95% CI
-4	8.11	No Data	180	14
3	8.06	No Data	204.94	0.01
10	8.17	0.01	203.69	0.04
17	8.16	0.02	203.78	0.04
31	8.14	0.01	217.07	0.04
38	8.15	0.01	212.60	0.04
45	7.26	0.01	209.54	0.04
52	7.25	0.01	207.82	0.04

Fig. 1 pH and alkalinity measurements from each day measured with 95% confidence intervals.

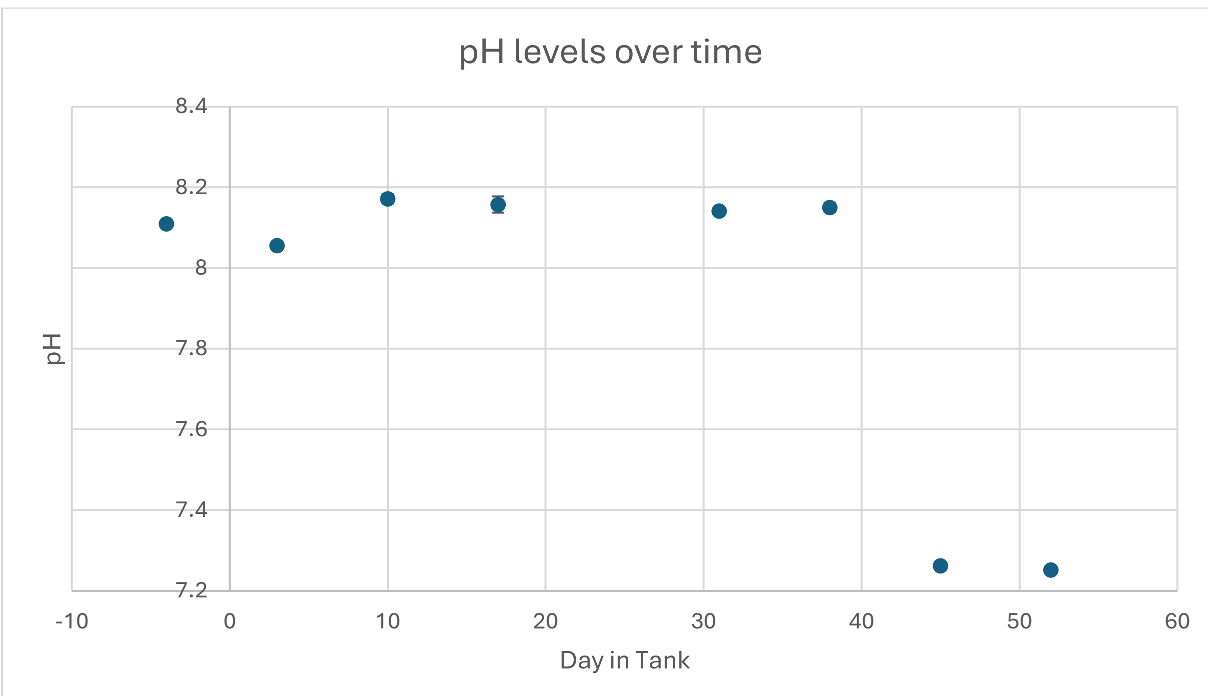


Fig. 2 pH levels plotted over day in the tank.

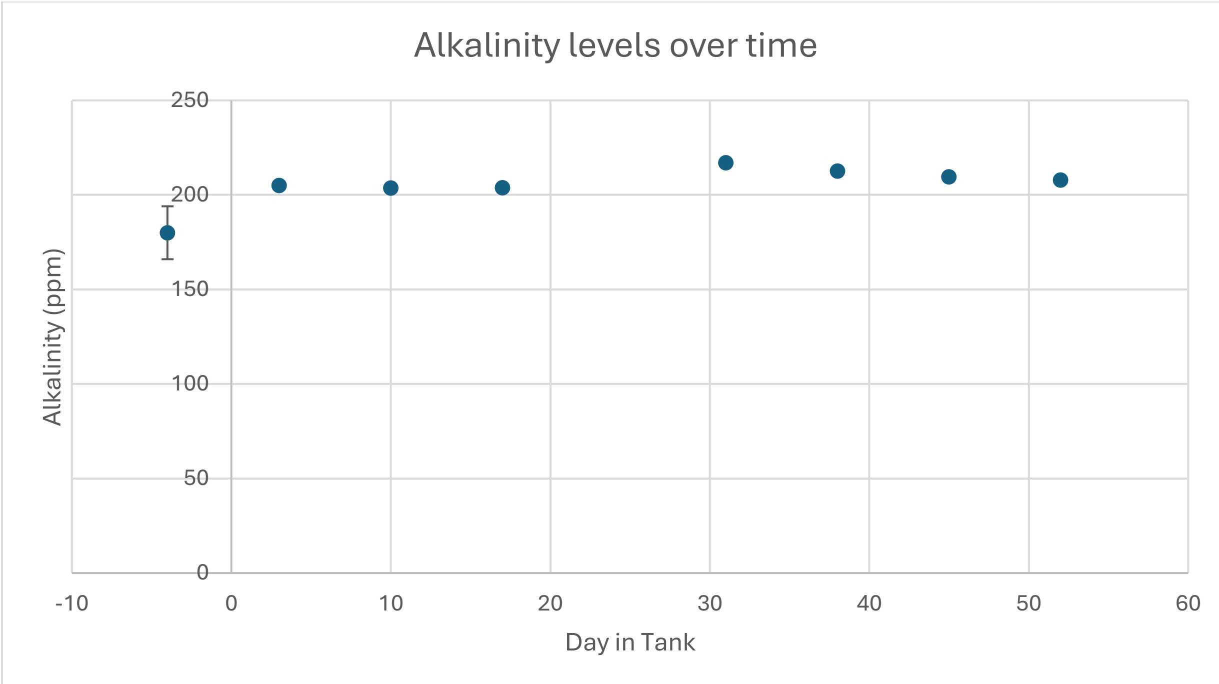


Fig. 3 Alkalinity levels plotted over day in the tank.

Discussion

Based on Fig. 3, it seems that alkalinity does not change much with time. Using a Grubb's outlier test, I determined that 180 is an outlier in the dataset, with a p-value of 0.007. While this could indicate a large change in alkalinity between day -4 and day 3 in the tank, it could also be a result of procedural error. If alkalinity increase between day -4 and day 3, it would make sense for pH to increase as well, since increased alkalinity indicates more basic ions in solution. However, according to Fig. 2, the pH actually decreases from day -3 to day 4. Therefore, given that this measurement was taken when the procedure was first developed, it makes sense that the value obtained on day -3 is a result of procedural error.

The largest change in alkalinity occurs between day 17 to 31. A large change in alkalinity could indicate a change in pH value, or other related analytes. Looking at day 17 and 31 measurements of pH, the 95% confidence intervals overlap, indicating that there is not a significant change in pH across day 17 and day 31. Alkalinity is also closely related to calcium and magnesium concentrations, so to hypothesize a reason for the alkalinity to change, it makes sense to see if there is a corresponding change in calcium and magnesium concentrations. Since there is no data for calcium and magnesium concentration on day 17, I will compare the change in alkalinity and the change in calcium and magnesium from day 10 to day 31 instead.

Day	Calcium w/ 95% CI (ppm)	Magnesium w/ 95% CI (ppm)	Alkalinity w/ 95% CI (ppm)
10	306 ± 0	1418 ± 133	203.68 ± 0.04
31	361 ± 27	1458 ± 66	217.07 ± 0.04

Fig. 4 Calcium, magnesium, and alkalinity measurements from day 10 and day 31.

Based on the 95% confidence intervals, it is clear that there is a significant increase in calcium concentration, but not a significant increase in magnesium concentration. We expect calcium concentration to increase as carbonate concentration increases because calcium carbonate exists in an equilibrium between the solid state and the dissolved state. An increase in carbonate ions indicates that more calcium carbonate has dissociated, increasing the amount of calcium in solution as well.

Looking at Fig. 2, pH seems to be relatively stable, and all changes are within the respective 95% confidence intervals. However, the last two points corresponding to the data collected on day 45 and 52 show a significant drop in pH. These values are nearly an entire pH level lower than the other vales in the dataset. In the spectrophotometric absorbances of these data points, the ratio of the protonated form of m-cresol purple to the deprotonated form was about 3 to 1. In other data points, it was the inverse, with about 3 times as much absorbance from the deprotonated form than the protonated form of m-cresol purple. This can be a result of incorrect notation, or a sudden change in pH in the water.

Day	578nm (Absorbance)	434nm (Absorbance)	578nm:434nm ratio
3	0.937 ± 0.001	0.4303 ± 0.0002	2.177
10	0.65 ± 0.09	0.24 ± 0.04	2.690
17	0.7 ± 0.1	0.27 ± 0.07	2.618
31	1.09 ± 0.05	0.43 ± 0.02	2.538
38	0.66 ± 0.05	0.26 ± 0.02	2.582
45	0.17 ± 0.03	0.42 ± 0.09	0.396
52	0.43 ± 0.06	1.11 ± 0.16	0.388

Fig. 5 Each absorbance value is the average of the 3 absorbance values from the given day's data set with 95% confidence intervals. Day -4 was omitted because there were not multiple trials run on that day, so confidence intervals could not be calculated.

To determine if the pH did change in seawater during day 45 to 52, we can look at other analytes related to pH to see if the change in their concentrations reflects the change in pH concentration. In seawater, the alkalinity, calcium and magnesium ion concentration, and ammonia concentration are the most closely related to pH. By comparing the average concentration of these three analytes before day 45 and then after day 45, we can determine if pH did change at day 45.

If pH were to drop significantly, there should be a corresponding change in carbonate concentration. In a more acidic solution, there will be less carbonate measured, as more of the carbonate will be protonated to form bicarbonate. To determine this, we can separate the data into two categories: alkalinity of the seawater when pH is around 8.1, and alkalinity of the seawater when pH is around 7.3. By performing a t-test on the mean of each set of alkalinity values, we can determine if there is a significant difference between the mean to a 95% confidence level.

Mean Alkalinity Days -3 to 38	**Mean Alkalinity Days 45 to 52**	**t-test p-value**
203.6773	208.6807	0.3866

Fig. 6 The mean of the two groups based on pH level and the p-value of the t-test to compare their means. The null hypothesis is that there is no statistical difference between the two means.

Since the p-value of the t-test is greater than 0.05, then the null hypothesis is retained, and there is no significant differences in the mean alkalinity between the two groups.

Calcium and magnesium ions interact with carbonate ions to form either calcite or aragonite, two polymorphisms of calcium carbonate. When the Mg^{2+} to Ca^{2+} ratio exceeds 5.2, the magnesium ions compete with the calcium nucleation sites. This forms aragonite, which is the more soluble form of calcium carbonate [7].

$$Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3$$

Fig. 7 Chemical equilibrium equations for calcium carbonate formation.

If the pH of the solution is lowered, then there should be less carbonate in solution, as carbonate is protonated to form bicarbonate. This will result in more calcium carbonate dissolving to make up for the lack of carbonate in solution. This will then lead to more calcium ions in the seawater as well. We can use another t-test to evaluate if the average amount of calcium ions in tank days 45 and 52 are statistically different from the average amount of calcium ions previously.

Mean [Ca^{2+}] Day -4 to 38	**Mean [Ca^{2+}] of Day 52**	**t-test p-value**
375 ppm	370 ppm	0.7621

Fig. 8 The mean of the [Ca^{2+}] of both groups and the p-value of the t-test to compare the two values. The null hypothesis is that there is no statistical difference between the two means. Since p-value > 0.05, the null hypothesis is retained.

Since the p-value of the t-test is greater than 0.05, then the null hypothesis is retained, and there is no significant difference in the [Ca^{2+}] between the two groups.

Ammonia concentrations are also pH-dependent. Ammonia in seawater exists in equilibrium with ammonium, where ammonium concentrations rise as pH lowers. In general, ammonium is considered to be less toxic than ammonia in seawater [8].

$$NH_3 + H^+ \rightarrow NH_4^+$$

Fig. 9 The equilibrium between ammonia and ammonium. As pH is lowered, the increase in hydronium ions will result in more ammonium being formed.

The same method used in Fig. 8 can be used to determine if the change in pH creates a statistically significant difference in ammonia concentration.

Mean $[NH_4^+]$ Day -4 to 38	**Mean $[NH_4^+]$ Day 45 to 52**	**t-test p-value**
0.382	0.039	0.02372

Fig. 10 The mean of the $[NH_4^+]$ of both groups and the p-value of the t-test to compare the two values. The null hypothesis is that there is not statistical difference between the two means, Since the p-value is < 0.05, then the null hypothesis is rejected.

The p-value of the t-test comparing $[NH_4^+]$ between the two groups is less than 0.05, so the null hypothesis is rejected, and there is a statistical difference between the ammonium concentrations between the two days. To determine which group had a higher concentration of ammonia, a one-tailed t-test is needed.

Mean $[NH_4^+]$ Day -4 to 38	**Mean $[NH_4^+]$ Day 45 to 52**	**One tailed t-test p-value**
0.382	0.039	0.01186

Fig. 11 The mean of the $[NH_4^+]$ of both groups and the p-value of the one-tailed t-test to compare the two values. The null hypothesis is that one of the means is not greater than the other. Since the p-value is less than 0.05, then the null hypothesis is rejected.

Since the p-value of the one tailed t-test is less than 0.05, then the null hypothesis is rejected, and the mean ammonium concentration from day -4 to 38 is significantly larger than the ammonium concentration from day 45 to 52. This does not line up with the expectation that as pH decreases in day 45 to 52, ammonium concentrations should increase. As a result, it is reasonable to assume that there is either an error in the ammonium data, or there is an error in the pH data.

Through evaluating the effect of the drastic pH change on carbonate, calcium, and ammonia concentrations, it is clear that the change in pH had no effect or the opposite expected effect on the concentrations of the analytes. Therefore, it is reasonable to assume that there is an error in the pH measurements on days 45 and 52, and that through further measurement, these 2 points will likely be shown to be outliers.

As shown in Fig. 4, the ratio of the absorbances measured on days 45 and 52 are inverses of the ratio measured on the other days. A likely explanation could be an error in writing down the data, leading to incorrect absorbance values and pH measurements. If the 578nm absorbance and 434 absorbance values were reversed, the resulting ratio would be similar to the other ratios obtained in earlier days. However, in order to be conclusive, further measurements should be taken to show that the pH remains around 8.2 and never deviates down to 7.2.

Conclusion

In conclusion, while alkalinity levels remain relatively stable, the largest change in alkalinity is reflected by an equally large increase in calcium ions, since calcium carbonate exists in equilibrium between the solid state and dissolved state. However, since alkalinity represents the buffering capacity of the seawater, and the pH of the seawater remained similar, there were no harmful results. Then, through analyzing the difference in concentrations of different pH dependent species, it is determined that the drastic change in pH did not create statistically significant changes in concentrations of other species. As a result, it is reasonable to determine that these values must be a result of procedural error, and that further testing of pH levels beyond day 52 will show that these 2 values are outliers in the data set.

Procedure

Required Glassware/Equipment

- 50mL volumetric flask
- 3 250mL beakers
- 10mL volumetric pipette
- 50mL volumetric pipette
- 3 magnetic stir bars
- Magnetic stir plate
- Titration burette setup
- 2 1cm cuvettes

- UV-Vis spectrophotometer
- Disposable pipettes

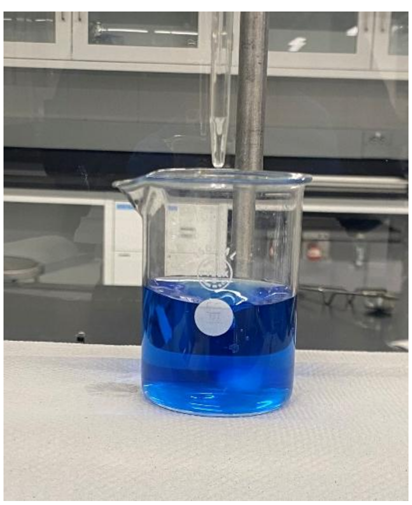
Part 1: Calibrate HCl stock solution

First, you must perform a titration to measure the exact concentration of a stock solution of approximately 0.07M hydrochloric acid.

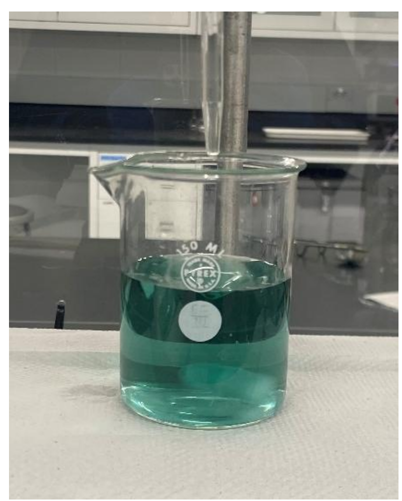
To start, create a solution of TRIS. TRIS molecular weight is 121.14g/mol. Weigh out by difference precisely 0.1276g of TRIS solid. Be sure to record the exact value. Dilute the TRIS solid to exactly 50.00mL. Make sure to swirl the solution in the flask to ensure that all the TRIS solid particles are fully dissolved in the water. Add exactly 10.00mL of the TRIS solution to a 250mL beaker. You will need to see the color of the solution to determine the titration equivalence point, so add about 20-30mL of deionized water in order to be able to fully see the color change. Add 4-5 drops of bromothymol blue (or until clear visibility) and a magnetic stir bar to the beaker. Bromothymol blue has a green equivalence point and is yellow in acidic solutions and blue in basic solutions. The solution should now be a dark blue color.

Set up a burette with the unknown concentration of HCl solution. Make sure to properly wash and rinse the burette with deionized water and then the hydrochloric acid solution. Place a magnetic stirrer under the burette and then the beaker of TRIS solution on top. For better visibility of the color change, place a white paper towel between the beaker and the magnetic stirrer so that the color of the solution can be clearly viewed against the white background.

Using this setup, titrate the HCl solution around 3 times. The end of the titration should be when the solution is a neutral green color (not blue green, but fully green). For each titration, you should use around 30 mL of HCl solution. Using the volume of HCl solution used and the known concentration of the TRIS solution, evaluate the true concentration of the HCl solution. If the standard deviation of the 3 derived concentrations of HCl are not within 5 percent of the mean, repeat the procedure until more accuracy is obtained.



Before titration:



After titration:

the solution should be a vibrant blue *the solution should be a neutral green*

Part 2: Titrate the seawater with the now known concentration of HCl solution

Next, you will titrate the seawater using the stock HCl solution. This titration will tell you the overall concentration of carbonate and bicarbonate in the solution.

Use a clean beaker to scoop some seawater out of the tank. Using a volumetric pipette, measure exactly 50.00mL of seawater into a clean 250mL beaker. Add 4-5 drops of bromocresol green to the seawater. Bromocresol green has an equivalence point around 3.8-5.4 pH. At this pH, both bicarbonate and carbonate will be fully titrated into carbonic acid. Bromocresol green is yellow in acidic solutions and blue in basic solutions. Around its equivalence point, similar to bromothymol blue, it will be a neutral green color. Titrate the seawater solution with HCl until this green equivalence point is reached. You should use around 30mL of HCl solution.

Repeat this titration around 3 times. Then, using the amount of HCl used and its known concentration, calculate the total amount of hydronium that was added to the solution. This value corresponds to the total amount of carbonate and bicarbonate ions in the solution.

HCl strong acid reaction: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$

Total hydronium ions: $[H_3O^+] = [HCO_3^-] + 2[CO_3^{2-}]$

Calculate the mean and standard deviation of all three trials. If the standard deviation is not within 5 percent of the mean, repeat the procedure until more accuracy is obtained. This will be the value will later on be used in the Henderson Hasselbach equation to determine the individual concentrations of carbonate and bicarbonate in the seawater.

Part 3: Evaluate the overall pH of seawater with spectrophotometric determination

In this step, you will use *m*-cresol purple dye and spectrophotometer to determine the overall pH of the solution.

m-cresol purple dye is present in solution in 3 forms: I^{2-}, HI^-, H_2I. These ions absorb at different wavelengths: 578nm, 434nm, and 410/510nm respectively. The m-cresol purple dye is originally in the I^{2-} form, where it is not protonated at all. When added to a solution, it will either stay in the I^{2-} form or protonate into one of the two other forms. By measuring the absorbance of the solution at various wavelengths, you can determine the concentration of each ion, then use Beer's law substituted into the Henderson Hasselbach equation to measure the concentration of hydronium in the solution. The concentration of hydronium in the solution will then allow you to evaluate the total pH of the solution.

To use the spectrophotometer, power up the spectrophotometer and click on SimpleReads application. Click on "Setup" and input the correct wavelengths separated by a comma. For the first measurement, fill a 1cm cuvette about 2/3 with water and click "Zero" to calibrate the instrument. Then, take a 1cm cuvette and fill it about 2/3 with seawater. Add 3 drops of *m*-cresol purple dye and pipette up and down with a clean disposable pipette to completely mix the dye into the seawater. Place this cuvette into the spectrophotometer, with the cloudy sides vertically oriented and the clear sides horizontally oriented. Close the lid of the spectrophotometer and hit start. Record the absorbances at each of the 3 wavelengths.

These wavelengths should be 578nm, 434nm, and 710nm (This last wavelength is a wavelength that the dye doesn't absorb at. Use it to subtract any background noise from the other wavelengths). Run about 3 trials to ensure that the readings are consistent. For ideal accuracy, the peak absorbances should be between 0.5-1A. If the peaks fall below this range, add one more drop of m-cresol purple dye and repeat the procedure. If the peaks are above this range, repeat this procedure with a clean cuvette and add 1 less drop of dye. It is okay if one of the peaks does not show up, this simply means that one particular form of the m-cresol purple dye was not present in the solution. Knowing that both carbonate and bicarbonate are bases, it is likely and expected that only the I^{2-} and HI^- species will be present in the solution.

Using these absorbance values, you will calculate the overall pH of the solution. The Beer's Law equation calculates the absorbance based on concentration, molar absorptivity, and length of light path, which is just the width of the cuvette.

$$\text{Beer's Law:} \qquad A = \varepsilon b C$$

This law can be manipulated to be concentration in terms of the other variables (derivation in appendix). Using this value of concentration, it can be substituted into the Henderson Hasselbach equation to produce:

$$\text{Manipulated H-H equation: } pH = pK_2 + \log_{10}\left(\frac{A_1/A_2 - \varepsilon_1(HI^-)/\varepsilon_2(HI^-)}{\varepsilon_1(I^{2-})/\varepsilon_1(HI^-) - (A_1/A_2)\varepsilon_2(I^{2-})/\varepsilon_2(HI^-)}\right)$$

A_1 is the absorbance at 578nm, corresponding to I^{2-}, and A_2 is the absorbance at 434nm, corresponding to HI^-. pK_2 is the acid dissociation constant for I^{2-} and HI^-. This can be calculated as a function of temperature and salinity.

$$pK_2 = \frac{1245.69}{T} + 3.8275 + 0.0021(35 - S)$$

$$293 \leq T \leq 303 \text{ and } 30 \leq S \leq 37$$

Temperature should be in Kelvin and salinity should be obtained from the board.

The extinction coefficients for m-cresol purple are as follows:

$$\varepsilon_1(HI^-)/\varepsilon_2(HI^-) = 0.00691$$

$$\varepsilon_1(I^{2-})/\varepsilon_2(HI^-) = 2.2220$$

$$\varepsilon_2(I^{2-})/\varepsilon_2(HI^-) = 0.1331$$

Using these values and the overall equation above, you can evaluate the overall pH of the solution.

Calculations:

After these experiments, you have an overall pH of the solution and a total concentration of bicarbonate and carbonate. To determine the alkalinity of the solution, recall that alkalinity is measured in ppm of hypothetical calcium carbonate it would take to reach the same amount of alkalinity. Therefore, the number of moles of hydrochloric acid divided by two is equal to the moles of carbonate ions in solution that it takes to reach the same alkalinity. Convert this value from moles to parts per million.

Using the Henderson Hasselbach equation, you can determine the individual concentrations of bicarbonate and carbonate through algebraic manipulation.

Henderson Hasselbach equation: pH = $pK_2 + \log_{10}\left(\frac{[CO_3^{2-}]}{[HCO_3^-]}\right)$

Note that the pK_2 of this equation is not the same as the pK_2 of the earlier equation. This is the pK_2 of carbonic acid. You can get this value of pK_2 by looking it up online or in the textbook.

Input the obtained pH, alkalinity in ppm, bicarbonate concentration, and carbonate concentration into the spreadsheet.

Appendix

Making stock HCl solution

The goal is to make a 1 liter of HCl solution at approximately 0.0071M. The lab stock solution is usually 12M. Add exactly 1mL of 12M HCl solution to a 10mL volumetric flask, then fill remaining volume with deionized water. Swirl to ensure all HCl is dispersed throughout the solution. Using a volumetric pipette, remove exactly 5mL of the HCl solution and then add to a large beaker. Add about 995mL of deionized water to create 1L of solution.

Derivation of Beer's Law substituted into Henderson Hasselbach equation

This derivation requires the ratio of absorbances measured at 2 different wavelengths. The first wavelength will be wavelength 1 and the second will be wavelength 2.

At λ_1, $\quad A_1 = \varepsilon_1(HI^-)[HI^-] + \varepsilon_1(I^{2-})[I^{2-}]$

At λ_2, $\quad A_2 = \varepsilon_2(HI^-)[HI^-] + \varepsilon_2(I^{2-})[I^{2-}]$

I will divide these equations, then isolate $\frac{[I^{2-}]}{[HI^1]}$.

$$\frac{A_1}{A_2} = \frac{\varepsilon_1(HI^-)[HI^-] + \varepsilon_1(I^{2-})[I^{2-}]}{\varepsilon_2(HI^-)[HI^-] + \varepsilon_2(I^{2-})[I^{2-}]}$$

$$\frac{A_1}{A_2}(\varepsilon_2(HI^-)[HI^-] + \varepsilon_2(I^{2-})[I^{2-}]) = \varepsilon_1(HI^-)[HI^-] + \varepsilon_1(I^{2-})[I^{2-}]$$

$$\frac{A_1}{A_2}\left(\varepsilon_2(HI^-) + \varepsilon_2(I^{2-})\frac{[I^{2-}]}{[HI^-]}\right) = \varepsilon_1(HI^-) + \varepsilon_1(I^{2-})\frac{[I^{2-}]}{[HI^-]}$$

$$\frac{A_1}{A_2}\varepsilon_2(HI^-) - \varepsilon_1(HI^-) = \varepsilon_1(I^{2-})\frac{[I^{2-}]}{[HI^-]} - \frac{A_1}{A_2}\varepsilon_2(I^{2-})\frac{[I^{2-}]}{[HI^-]}$$

$$\frac{A_1}{A_2}\varepsilon_2(HI^-) - \varepsilon_1(HI^-) = \frac{[I^{2-}]}{[HI^-]}\left(\varepsilon_1(I^{2-}) - \frac{A_1}{A_2}\varepsilon_2(I^{2-})\right)$$

$$\frac{\frac{A_1}{A_2}\varepsilon_2(HI^-) - \varepsilon_1(HI^-)}{\varepsilon_1(I^{2-}) - \frac{A_1}{A_2}\varepsilon_2(I^{2-})} = \frac{[I^{2-}]}{[HI^-]}$$

$$\frac{\frac{A_1}{A_2} - \frac{\varepsilon_1(HI^-)}{\varepsilon_2(HI^-)}}{\frac{\varepsilon_1(I^{2-})}{\varepsilon_2(HI^-)} - \frac{A_1}{A_2}\left(\frac{\varepsilon_2(I^{2-})}{\varepsilon_2(HI^-)}\right)} = \frac{[I^{2-}]}{[HI^-]}$$

Sources:

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